Practice Set: Minor II

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1. Write a program to do the following: Read a number a from standard input. If a is divisible by 4, print “X”. If a is divisible by 3 but not divisible by 4, print “Y”. If a is not divisible by 3 or 4, print “Z”.
2. Write a program to read characters from the standard input using gertchar() function till a ‘*’ is encountered; and print the number of times the character $ appears in the characters that are read.
   Example input: An# B CC g %f*aB$
   Output: “Number of $: 2”
3. Write a program to read two integer numbers a and b and print “X” if a – b is 1, “Y” if it is 2, “Z” if it is 3 and “O” otherwise. (**Hint:** Use switch statement)
4. Write a function that takes an integer array marks and the length of the array n as input and returns the product of the first and second elements if it has at least two elements; and 0 otherwise.
5. Write a function bool areStringsSame(char* a, char* b) that takes two character strings as arguments and returns true if they are the same, false otherwise.
6. Write a function bool match(char* a, char* b) that takes two character strings as arguments and returns true if they match in the first two characters, false otherwise. For example, match(abcd, abe) returns true.
7. What is a recursive function? Write a recursive function that takes integer n as argument and returns the sum of the first n numbers if n is a positive integer, 0 otherwise. (Done in class and also in the notes)
8. What is passing by value and what is passing by reference? How do you decide which one is needed for a function that you have to write?
9. Consider the function below. Write down the sequence of calls to and returns from f, in the correct order, when f(7) is called.

```
int f(int n)
{
    if (n <= 0)
        return -1;

    if (n == 1)
        return 0;

    if (n == 2)
        return 1;

    return (f(n - 1) + f(n – 2);
}
```

10. Write a function to swap the value of two variables. (Done in class)